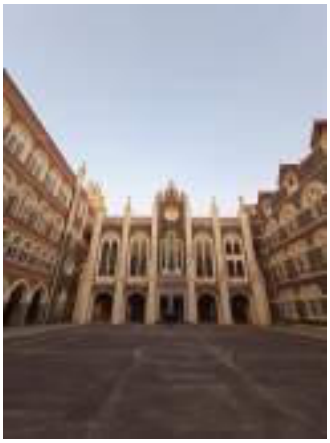
ST. XAVIER'S COLLEGE (AUTONOMOUS), MUMBAI
MA IN ANCIENT INDIAN HISTORY, CULTURE AND ARCHAEOLOGY
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PAPER AAIC0903- STUDIES IN CONSERVATION AND RESTORATION
(UNDER THE GUIDANCE OF - DR. JASON JOHNS)

TOPIC
"ST. XAVIER'S COLLEGE"
CONSERVATION ACTION PLAN



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INTRODUCTION

St. Xavier's College is a private, Jesuit educational institution located in the Fort district of the city of Mumbai, Maharashtra, India (18.9426° N, 72.8311° E). Founded and run by the Bombay Province of the Society of Jesus, a Christian religious organization. Affiliated with the University of Mumbai and granted autonomy by the same, the college stands as an important landmark not only in the realm of education but also in architectural heritage. Designed and built in the Indo-Gothic style of architecture and stretched on a 2.94 acres campus, the institution has been recognised and designated a "heritage structure" by the Mumbai Heritage Conservation Committee.

The college's architecture is a masterpiece, showcasing the Indo-Gothic style that seamlessly fuses Indian and European design elements. The campus, spanning 2.94 acres, provides an inspiring backdrop for learning and cultural exploration. Its meticulous design not only captures the spirit of the past but also retains its relevance in the present.

Earning recognition as a "heritage structure" by the Mumbai Heritage Conservation Committee, St. Xavier's College takes on the role of a guardian of history. This distinction underscores the institution's dedication to safeguarding its architectural and cultural legacy. Amidst a rapidly evolving urban landscape, the college's commitment to preservation stands as a beacon of continuity.

In essence, St. Xavier's College is a harmonious blend of education and heritage. Its architectural brilliance and recognition as a heritage structure highlight its dual role. As time moves forward, the college remains an unwavering symbol of reverence for history and a source of inspiration for generations to come.

HISTORICAL BACKGROUND

St. Xavier's College, named after Francis Xavier, a 16th-century Spanish Jesuit saint who incepted the educational endeavor of the Society of Jesus in India, was founded on January 2, 1869 in Bombay by German Jesuits. Saint Francis Xavier was a Spanish priest from Javier who came to the subcontinent during Portuguese dominion to impart the word of God to the inhabitants of India. He was one of the most influential Christian individuals in India. The brotherhood that founded the institution, on the other hand, was German in origin. Because it was a German institute in British India, the college faced broad repercussions during World War I (1914-1918), as Germany was considered as a rival force. Following the onset of the war, the German Jesuit priests, primarily the older ones, were incarcerated and detained at the Khandala College Villa in 1914, where many died. The departure of German Jesuits caused a schism in the college's administration, which was mitigated by a few Swiss, Luxemberger, and English Jesuits. Even after the war, German control of the institution was never fully restored, and other European Jesuits, and now Indian Jesuits, took over. The college is now led by Indian Jesuits, who place a strong emphasis on affirmative action for minority students.

The institution was originally located on Carnac Road in Fort, but was later relocated to its current location after the government offered the institute a plot of land for 999 years at a fee of Rs. 1 per annum. While its inception was as an arts college with only two students in attendance, it extended considerably and science departments were formed by the 1920s. Joseph Michael Willy, a Swiss Jesuit, was the first principal who was also a lecturer at the college and began teaching on 7 January 1869. For 43 years, only male students walked the regal halls of St. Xavier's College, until women were allowed to enroll in 1912. For the first time, women were admitted to the college in January 1912. There were just nine women in the inaugural group, and none of them continued their education at the institute. The second class featured 13 female students, three of whom graduated and the others dropped out. The university-affiliated college now offers undergraduate and postgraduate studies in Arts, Science, Commerce, and Management. In 2010, Xavier's became the University of Mumbai's first autonomous college. St. Xavier's received an 'A+' rating from the National Assessment and Accreditation Council (NAAC) in 2006[1].

[1]Assessment Report on Institutional Accreditation of St. Xavier's College, Mumbai, Maharashtra. (2007). National Accreditation Council (NAAC).

Architectural Developments:-

The architectural plan of both the exterior and interiors of the institution was executed in the classic Gothic Revival or Indo-Gothic style with a somewhat later addition in the Venetian Gothic Style in the main courtyard. The Indo-Gothic style, sometimes known as the Indo-Saracenic style, was a revivalist architectural style primarily adopted by British architects in India in the late nineteenth century, particularly in public and government buildings in the British Raj and the palaces of princely state rulers. St. Xavier's College is a prominent example of this architecture, as are many of the buildings in Old Bombay. It took aesthetic and ornamental aspects from native Indo-Islamic architecture, particularly Mughal architecture, which the British considered to be the traditional Indian style. The basic layout and structure of the structures tended to be similar to that utilized in current buildings in other revivalist styles, such as Gothic revival and Neo-Classical, with distinctive Indian characteristics and decoration added. Furthermore, varied architectural elements, from the towers and battlements suggestive of the Spanish style to some elements of the West Walton Church, Norfolk can also be found interspersed throughout the institution making it easy to discover the inspirations.

The college witnessed massive architectural expansions from 1884 to 1910 under the patronage of Principal Frederick Drekmann and from 1933-1940 under Father G. Palacios during his tenure as the principal, the latter having added a new hall, the central library and the chapel. The library was built by Fr. G. Palacios to be situated in the center of the college complex, above the hall and below the classrooms, with the new chapel on one side and the lecture halls on the other. It also dominates the front quadrangle on one side and the tennis court on the other. The library's doors and catalog chests were carved in the manner of the early German artisans.

Following the withdrawal of the German Jesuits, the number of lay professors grew. Enrollment of students from outside Bombay, particularly Calcutta, Yangon, Mangalore, and Sindh, began to increase in 1920. As a result, a third storey was erected to the hostel to supply them with lodging. Leslie Orme Wilson, Governor of Bombay (1923-1926), inaugurated the east-west scientific wing addition in 1925. The college hostel was finished in 1909, and the east-west scientific wing was finished in 1912. The government offered grants of Rs. 70,000 and Rs. 37,000 for the college's two additional structures. Ethelbert Blatter, a Swiss Jesuit priest, and his associates founded the Blatter Herbarium in 1906. It is India's largest botanical library, preserving over 200,000 specimens of algae, fungus, bryophytes, flowering plants, and other organisms.

The college chapel was constructed under the direction of Father G. Palacios and has certain elements of perpendicular Gothic. Thick teak wood panels with carvings of human values adorn the three doorways at the chapel's base. Each quality is emphasized by a carving of a famous person. Leonardo da Vinci represents the spirit of inquiry, King Emperor George VI stands for authority, Mughal Emperor Akbar represents splendor, General Francisco Franco stands for leadership; Vasco Nez de Balboa represents business; Esther, the Jewish Queen of Assuerus represents beauty; Samson represents strength; Dorabji Tata represents wealth; Louis Pasteur represents science; Michelangelo represents art; St. Isidore of Seville represents work; St. Elizabeth of Hungary represents love; Rabindranath Tagore represents kindness; Solomon represents wisdom; and Cardinal Francisco Jiménez de Cisneros represents statesmanship. The college chapel's construction, which began in 1937, was finished under the administration of Principal Aloysius Coyne (1940-1949). Lord Brabourne, Governor of Bombay from 1933 to 1937, inaugurated the college hall in January 1937.

The non-degree course for the Teacher's Diploma began in August 1939, and Microbiology was introduced. Following India's independence in 1947, Hindi began to be taught in the college in June 1949, and numerous new departments were established, including the Departments of Sociology and Anthropology (1951) and Psychology (1957). St. Xavier's College continued to grow in student body and staff size in the late twentieth century, and several research institutes, programs and departments were established. The institution has been managed by Indian Jesuits from the Bombay Province in close collaboration with the Society of Jesus in Germany and Spain.

EXAMINATION AND DOCUMENTATION

St. Xavier's College, Mumbai is a heritage building standing as a proud testament of an Indo-Gothic architectural heritage. Preserving these structures requires meticulous examination and documentation to ensure their longevity and accessibility for future generations. The use of advanced tools such as distometers, Dino lights, UV lights, GPS, compasses, epicollect5 app and Google Earth has revolutionized the way conservators approach this task.[2] The examination and documentation process is the cornerstone of a successful conservation action plan. It involves detailed assessment of a building's physical condition, materials, and historical context. Advanced tools have greatly facilitated this process, allowing conservators to capture intricate details that might have been overlooked using traditional methods.[3]

- Visual Inspection based on eye level

  A thorough observation of the St. Xavier's College site was conducted to identify the potential hazards to the heritage structure. The prime places included the 1st Quadrangle, the chapel, the terrace and the expansion joint connecting the Hogwarts staircase structure to the building. Based on the visual inspection, samples were collected from these locations for further analysis.

- Visual Inspection based on lights and radiation

  Dino lights are digital microscopes, offering an unprecedented look into the minute details of a building's surfaces and materials. These tools utilize high-resolution imaging to reveal hidden features and deterioration that might not be visible to the naked eye. By examining the building's surfaces at a microscopic level, conservators can identify structural weaknesses, assess the extent of damage, and make informed decisions about necessary interventions.

  UV lights play a crucial role in identifying areas of the building that have been subject to alterations or restoration work over time. Different materials fluoresce under UV light in

[2] Tiňo, R., Vizárová, K., Krčma, F., Reháková, M., Jančovičová, V., Kozáková, Z. (2021). Plasma Technology in the Preservation and Cleaning of Cultural Heritage Objects. United States: CRC Press.
[3] Mezzino, D., Chan, L., Quintero, M. S., & Espondo, M. (2017, August). *Built Heritage Documentation and Management: An integrated conservation approach in Bagan*. researchgate.

distinct ways, aiding in distinguishing original components from later additions. This information ensures that restoration efforts remain faithful to the building's original design and historical significance.

- Metrics / Measurements / Locations

  Distometers are integral to measuring accurate distances, heights, and angles of various architectural elements. These handheld devices use laser technology to provide precise measurements, ensuring that restoration work is executed with unparalleled accuracy. By assessing dimensions down to the millimeter, conservators can recreate intricate details that might have otherwise been lost.

  Measuring tapes are manual instruments utilized to measure small distances. It is a flexible and portable tool used for measuring distances or lengths. Measuring tapes come in various lengths and units, such as inches, feet, centimeters, or meters, to suit different measurement needs.

  Compasses continue to be invaluable tools for heritage building examination, aiding in determining the alignment of structures. By understanding the building's alignment with geographical features or cardinal points, historians and conservators can gain insights into the cultural, religious, and functional aspects of the structure's design[4].

  Global Positioning System (GPS) is a satellite-based navigation system that allows users to determine their geographical location anywhere on Earth. GPS provides location coordinates in the system. It is highly useful in combination with other techniques and appropriate in documenting mass targets and structural deformation[5].

- Monitoring Equipment

  Camera is a device used to document and record visual images of a heritage site. In heritage documentation, the choice of camera equipment, including the type of camera, lenses, and accessories, is crucial to ensure the accuracy and quality of the visual records. Mostly, SLR and phone cameras were utilized to document the site.

[4] Kovacs, Fereshteh. (2015). Documentation of cultural heritage; techniques, potentials, and constraints. ISPRS - International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences.
[5] Ibid

Grid lines refer to a pattern of horizontal and vertical lines that are superimposed on an image or document to help with alignment, positioning, or measurement. The SLR and phone cameras have in built grid lines that assist in alignment,

- Softwares

EpiCollect5 is a free and open-source application designed for data collection and field surveys. It is particularly useful for researchers, conservators and individuals who need to collect, document, manage, and analyze data of sites. The application collects and organizes data in words, photographs, audios and videos. It automatically detects the date, time, latitudes and longitudes. EpiCollect5 was developed by researchers at Imperial College London.

Google Earth, a geographical and spatial tool, adds a new dimension to heritage building examination and documentation. By overlaying historical maps and images onto modern satellite imagery, conservators can trace the evolution of the building and its surroundings over time. This aids in understanding the building's context, historical development, and the impact of urbanization on its preservation.

The integration of these advanced tools in examination and documentation has transformed heritage building conservation into a multi-faceted, data-rich endeavor. However, it's crucial to note that these tools do not replace traditional methods but they rather compliment them. Expert knowledge and on-site inspections remain essential for interpreting the data collected through these tools, as they provide valuable insights required for effective decision-making[6].

[6] Conservation of Cultural Heritage. Test Methods. Measurement of Water Absorption by Pipe Method. (1913). United Kingdom: B S I Standards.

SAMPLE ANALYSIS

Sample 1

Material- Tile
Location- 1st quad, outside seminar room

Examination under dino light revealed yellowish granules of silica on the sample. A minor experiment involving hydrochloric acid revealed the nature and type of stone used in the manufacturing of the tile to be sandstone.

Sample 2

Material- Plaster
Location- 1st quad, outside seminar room

Examination under dino light highlighted tiny granules of dust, hair and varied foreign material on the sample.

Sample 3

Material- Plaster and paint
Location- Expansion joint, lr 23

During examination under dino light, the painted portions of the plaster appeared glossy whereas the platser showcased a matte finish. Dust, hair was also observed on the sample.

Sample 4

Material-Cement
Location- Expansion Joint, lr.23

Examination under dino light revealed the sample to be a hydraulic type of cement. The sample was covered with dust, fiber and lint particles.

Sample 5

Material- Mold and algae
Location- Terrace, outside the wall of MVM office

The painted side of the sample showed mold growth while the other side had an algae or moss accumulation. Inspection under dino light also showed brown spots which were caused due to expansion of dried up algae. The sample was also inspected under UV light which highlighted mold and algae, appearing dark and dark yellow in color respectively.

Sample 6

Material -Micro-vegetation
Location - Terrace outside wall of MVM office

Inspection under dino light revealed the sample to be funaria, a type of moss which grows in dense patches or cushions in moist shady and cool places on rocks or walls during the rainy seasons.

Sample 7

Material- Paint
Location -Terrace washroom door behind lr 44

Examination under dino light suggested microbiological decay due to presence of algae growth on the sample

Sample 8

Material-Nail
Location - Terrace, heap of stone structure near solar power plant

Examination of the iron nail under Dino light showed signs of decay in the form of corrosion due to oxidation of iron.

Sample 9

Material- Stone
Location-terrace, heap of stone structure near solar power plant

Inspection under dino light revealed growth of algae inside the stone. Pink and orange spots were also examined which were brick powder. Sand was accumulated on the sample in the form of crystals. UV light was also used to examine the sample, which confirmed the presence of algae, appearing dark yellow in color.

Sample 10

Material- tar
Location-terrace, heap of stone structure near solar power plant

Examination under dino light revealed the sample to be a petroleum product coated on paint. The sample also showed micro algae accumulation. Under UV light the algae appeared dark yellow in color.

Sample 11

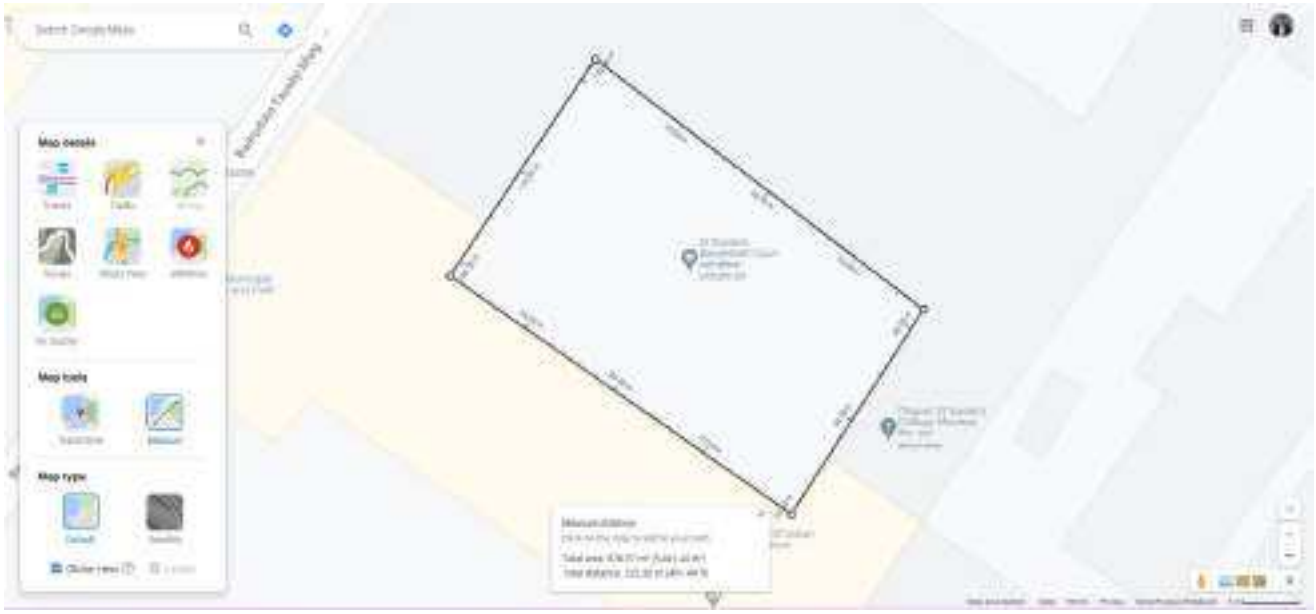
Material-brick
Location-terrace, heap of stone structure near solar power plant

Inspection under dino light revealed that the sample contained clay, sand (white particles), lime particles and blue color plastic particles. The sample also showed micro vegetation mostly moss. Refer to plate no.2

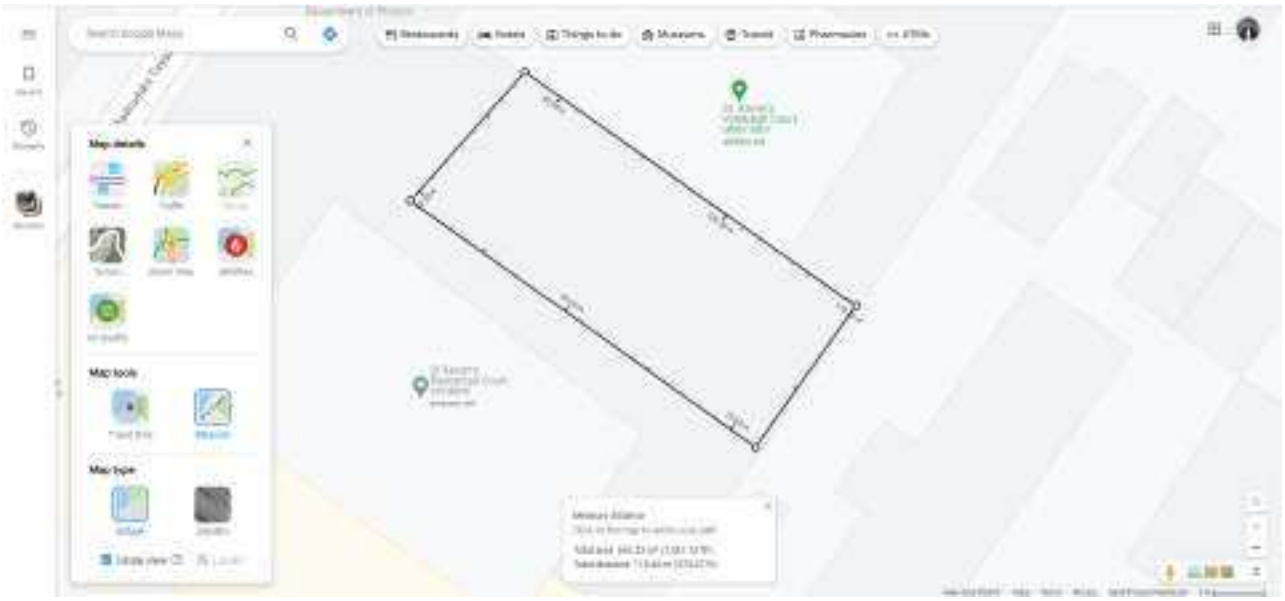
Examination and documentation is a crucial step of conserving our invaluable heritage buildings. The use of advanced tools such as distometers, Dino lights, UV lights, compasses, and Google Earth provide the precision, efficiency, and comprehensiveness for the process. These tools enable conservators to delve into the intricate details of a building's design, materials, and history, guiding restoration efforts with accuracy. However advanced the tools, the human touch of expert interpretation and on-site analysis remains irreplaceable in fully understanding and preserving the essence of these architectural marvels.

MAPS OF THE AREA EXAMINED:-

1st Quadrangle:-



Terrace:-



MATERIAL AND TECHNOLOGY

The Materials and their Chemical Composition utilized in the construction of St. Xavier's College.[7]

- Basalt - 45-52 wt.% Silicon dioxide (SiO2), 2-5 wt.% total alkalis, 0.5-2.0 wt.% Titanium dioxide (TiO2), 5-14 wt.% Ferrous oxide or Iron (II) oxide (FeO) and 14 wt.% or more Aluminum oxide (Al:O), 10 wt.% Calcium oxide (CaO) and 5 to 12 wt.% Magnesium oxide (MgO). The stone masonry on the chapel building, library wall, entrance wall and lab building are constructed with basalt.[8]

- Rhyolite - 70-75% Silicon Dioxide + Potassium Oxide ($SiO2 + K_2O$) It is a yellow ochre colored stone used on the lab building and library wall.

[7] Sethna, S.F.. (1999). Geology of Mumbai and surrounding areas and its position in the Deccan Volcanic Stratigraphy, India. Journal of the Geological Society of India. 53. 359-365.
[8] Naik, Anmol & Sheth, Hetu & Samant, Hrishikesh & Dsouza, Shawn. (2018). Some observations on spilitized basalts of the western Deccan Traps, India.

- Limestone - Calcium carbonate (CaCO3). The first quadrangle is lined with limestone.

- Mortar - It is a mixture of Calcium hydroxide (Ca(OH)2), silica and water. Limestone mortar is used as binding material of masonry units in construction.

- Brick - It contains high amounts of Silicon dioxide (SiO2), Calcium oxide (CaO) and Aluminum oxide (Al2O3) and low amounts of Ferric Oxide Iron (III) oxide (FeO3). Magnesium oxide (MgO), Potassium oxide (KO) and Sodium oxide (Na2O).

- Gypsum Plaster - Calcium Sulphate Dihydrate (CaSO.. H_2O) They are used for their adhesive properties and quick setting, used for construction.

- Metal - Metal, a multi-purpose material, used for beams, clock, window grills, etc.

- Wood - Wood is used on the windows, arches, windows and doors.

- Stained Glass - Stained or colored glasses are found on the chapel building.

GLOSSARY OF DAMAGES

Chapel

Sr. No.	Description of damage	Location	Factor of deterioration	Image
1.	Dirt accumulation	Wooden entrance door on 2nd floor of the Chapel building	Rh, Temperature	
2.	Discoloration/ water stains	Wall on left side of the wooden entrance door on second floor of the Chapel building	Rh, Natural phenomenon (rain/monsoon)	
		Wall on right side of the wooden door on second floor of the chapel building		

		Window wall on the right side of the wooden entrance door on second floor of the Chapel building		
3.	Flaking of plaster	Window wall on the left side of the wooden door on second floor of the chapel building	Accidents, Improper successive plaster coatings	
		Window wall on the right side of the wooden door on second floor of the chapel building		

4.	Cracks	Window on the left side of the wooden entrance door on second floor of the chapel building Window on the right side of the wooden entrance door on second floor of the chapel	Accidents	 
5.	External addition of wooden frame during repair	Outside Wall of the chapel passage	Vandalism Human Error	
6.	Chipping of paint	Outside ceiling of chapel	Neglect External forces RH and Temperature	

				
7.	Discoloration	Right side of the chapel entrance door	RH & Temperature Light Exposure	
8.	Cracks through the structure	Stairway leading to the upper section of the chapel	External forces Neglect	

				
9.	Broken pillar, Chipping of paint	Outside Window below the stairway	Neglect RH and Temperature Light	 

10.	Broken pieces of a structure	Outside the upper floor of the chapel entrance	Neglect	

Glossary of Damages - Chapel



Glossary of Damages - Chapel 2nd floor



Terrace

Sr. No.	Description of damage	Location	Factor of deterioration	Image
1.	Biodeteriotion	Northeast facing pillar on the left side of the Hindi Department	Water, Rh-temperature, Light exposure, Air pollution	

		Southwest facing pillar on the left side of MVM, behind Lr.41 Southern facing cubicle wall of the toilet behind Lr.44		 
2.	Micro vegetation	Heap of stonework in front of the north-west facing wall of the washroom beside Lr.44	Damp environment, Temperature, Light exposure	

		Below the door of southern facing cubicle wall of the toilet behind Lr.44		 
3.	Broken structure/missing apex	Northeast facing pillar on the left side of the Hindi Department Southwest facing pillar on the left side of MVM, behind Lr.41	Accidents, Neglect	  i

		Left pillar of the toilet behind Lr.44 Heap of stonework in front of the north-west facing wall of the solar power plant beside Lr.44		 
4.	Discoloration	Southern facing cubicle wall of the toilet behind Lr.44 Southwest facing pillar on the left side of MVM, behind Lr.41	Rh, Light exposure Air polluiton	

				
5.	Corrosion	Door of the southern facing cubicle wall of the toilet behind Lr.44	Rh-temperature	

6.	Cracks	Southwest facing pillar on the left side of MVM, behind Lr.41	Decay of building fabrics due to damp environment	
7.	Rusting	Base of the southern facing cubicle wall of the toilet behind Lr.44	Rh-temperature	
9.	Macro vegetation	Behind the southern facing cubicle wall of the toilet behind Lr.44	Rh-temperature	

Glossary Of Damages - Terrace



Expansion Joint

Sr. No.	Description of damage	Location	Factor of deterioration	Image
1.	Cracked, impaired cement and concrete floor	South east facing wooden door between Lr.24 and Lr.25, on the second floor	Poor workmanship, External Forces, Rh-temperature	
		South East facing wooden door at Lr.23 on the second floor.		

		South east facing wooden door right next to the Jesuits private quarters on the third floor.		
2.	Cracked wall	Above the south east facing wooden door at Lr.23 on the second floor.	Variations in temperature and humidity levels	
3.	Rusting	Lr.23 expansion joint behind the door on the right side	Rh-Temperature	

4.	Beehive	Wall rail near the south-east facing door on the 3rd floor.	Neglect	
5.	Tenting floor tiles	Next to the department of statistics, Lr 22	Moisture, Expansion or cracking of concrete below the tiles, Weak tile adhesive	

Glossary Of Damages- Expansion Joint



1st Quadrangle

Sr. No.	Description of damage	Location	Factor of Deterioration	Image
1.	Chipping of paint	North East Entrance arch of the passage leading to the volleyball court	Accidents (Truck passing through the passage) Weathering Bird Excreta	
2.	Beehive	Pillar 3 of the Hall Passage	Moisture Neglect	
3.	Water leakage	Pillar 4 of the Hall passage	Neglect Human Error	

				
4.	Crack in the pillar	Pillar 5 of the Hall passage	Accident Human Error	
5.	Crack and chipping of the stone pillar	Pillar below the chapel outside the Seminar room	Structural weight Neglect	
6.	Broken Limestone tiles on the edge of the Quad	Quad Boundary outside the seminar room	Human Error Neglect External forces RH & Temperature Light Exposure	

Glossary of Damages- 1st Quadrangle



CONSERVATION UNDERTAKEN IN THE PAST

The Chapel

Following interventions were found to have taken place.

- Additions and Architectural Conservation

Additional wooden beams were added to uphold the structural integrity of the corridor leading to the Chapel building. Though these beams were added to maintain the structural integrity of the passageway (corridor) it was still being affected with water damage as no necessary measures were undertaken to prevent water leakage. Due to the water damage present on the building it was noted that the plaster and mortar covering the brick structure was broken. Plate No. 9

The Expansion Building

- The concrete has spalled in the past and has been repaired in patches.The floor was repaired by the removal of spalled concrete and cement touch-up. However, the structural integrity of the floor/slab has degraded, its structural capacity might not fulfill the current codes.

The Terrace

- Addition of Solar Panels and Inverter room

In the last ten years, the terrace of the reference library building has undergone some renovations. In order to keep the college using green energy, solar panels and an inverter room for the solar panels were added. These solar panels were first installed in 2004 and have since undergone renovations and expansions. Due to age, some solar panels underwent an upgrade in 2019–2020. Originally a restroom, the inverter room was eventually converted into an inverter room for the solar panels. Although the terrace has undergone major renovations, fundamental maintenance like cleaning and upkeep should also be prioritized.

PRESENT CONSERVATION MEASURES

The Chapel

- Cleaning and Painting of the building

In the months of June and July, the Chapel building underwent cleaning. In order to maintain the Chapel's aesthetic value and give it a cleaner appearance, the interior of the building was also painted. Metallic wire brushes were used to clean the difficult corners of the Chapel. Even after taking these steps, the building's upper viewing galleries (Chapel 2nd floor) repainted walls still had apparent water damage, indicating that the requisite waterproofing precautions had not been done.

1st Quadrangle

- Plastering, painting and general cleaning of the area

1. In the recent months of 2023 the pillars of the Chapel building facing the 1st Quadrangle were painted and plastered. The sample of the chipped paint and plaster was examined under the Dino-lite (digital microscope).
2. Before the college fest (Malhar) began, one of the pillars facing the hall was vandalized (a piece of paper was taped to the wall). As soon as the festival was over, this piece of paper was removed.
3. The pictures in the glossary of damages for the 1st Quadrangle show that pillar 3 of the hall passage had a beehive attached to it. The team observed that this beehive was removed during the month of August.

RISK ASSESSMENT AND RECOMMENDATION FOR FUTURE CONSERVATION

Risk Assessment

6 major agents of deterioration to the structure of St. Xavier's college have been recognised. These include high relative humidity, temperature, light exposure, accidents, neglect/poor workmanship and air pollution.

Relative humidity is the amount of water vapor actually present in the air, expressed as a percentage of the maximum amount of water vapor the air can hold at the same temperature. Most of the time when it is raining the relative humidity is 90-99 percent[9] and relative humidity over 75 percent can cause different types of damages like speeding up of metal corrosion or bio-deterioration. Since St. Xavier's College (the structure under consideration) is situated in a region which receives heavy amounts of rainfall during the months of June to September, high relative humidity becomes a prime cause for multitude of the damages endangering the structure. Incorrect relative humidity and incorrect temperature work hand in hand as agents of deterioration since they affect each other and cohabitate. Temperature fluctuations or extreme temperatures can cause faster deterioration, deformation, dehydration, embrittlement etc. The rate of most chemical reactions, including deterioration, is approximately doubled with each increase in temperature of 10'C.[10]
Examination and documentation revealed that high relative humidity along with incorrect temperature were the cause of maximum amount of damages such as accumulation of dirt in isolated areas; water stains on walls and masonry; discolorations of walls and masonry; cracks on walls, masonry; chipping of paint off walls; corrosion; rusting; isolated micro and macro vegetation; and biodeterioration.

Light exposure -There are different kinds of lights that can cause various damages- visible light, UV and infrared light. These three forms of light have different damaging effects on objects. Ultraviolet rays have a very high energy and lead to yellowing, visible light causes fading or bleaching of colors and any kind of damage caused by infrared rays is due to exposure to heat.[11]

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[9] *Weather.gov > Louisville, KY > Discussion on Humidity*. (n.d.). National Weather Service. Retrieved August 27, 2023, from https://www.weather.gov/lmk/humidity
[10] Ogden, S. (n.d.). *2.1 Temperature, Relative Humidity, Light, and Air Quality: Basic Guidelines for Preservation*. NEDCC.
[11] Heitner, & Scaiano. (1993). *Photochemistry of Lignocellulosic materials*. American Chemical Society.

Masonry pillars and walls in the college have developed discoloration as a result of prolonged sun exposure (visible light) . Moreover, the triad of relative humidity, temperature and exposure to sunlight also provide favorable conditions for growth of micro vegetation such as funaria, moss, lichen on various isolated surfaces such as stone.

Accidents and external forces can cause considerable breakage, wear and tear, abrasion to heritage sites. Accidents often happen due to incorrect handling, during storage and transportation, construction works, overload etc. Moreover, multitude of people commute to the college daily, increasing risk of accidents causing damage to the structure. One such damage was documented at the college on the arch of the passage leading to the volleyball court. The arch exhibits two abrasion marks (at equal distance from the middle of the arch) which were caused as a result of entry of a vehicle through the passage. The vehicle was wider than the width of the arch and hence caused abrasion of the surface of the arch, while passing through it.

Poor workmanship can make the structure vulnerable to multitude of hazardous damages such as collapsing of structure, weakening of structure, inability of structures to withhold natural disasters like earthquakes etc. At St. Xavier's college, particularly in the expansion building, various evidence of poor workmanship have been documented. On the second floor, cracks running through the concrete floor can be spotted throughout. Attempts were made in the past to refill the cracks by the removal of spalled concrete and cement touch-up. However, the touch up has also developed cracks, thus weakening structural integrity of the floor. On the 3rd floor of the expansion building, tile tenting has been spotted. Basically lifting of a section of tiles due to improper application of tile adhesive during the laying of tiles. Poor workmanship disrupts the structural integrity of the structure making it vulnerable to multitude of damages

Pollutants contribute heavily to the deterioration of surfaces of buildings and monuments. Often associated with industrial and urban activities, air pollution can have a significant impact on stone materials used in buildings and structures, sulfur and nitrogen oxides are particularly harmful pollutants especially for carbonate stones (like limestone).[12] These oxides react with moisture in the air, resulting in the formation of acids which erode the stone surface and gradually compromise its structural integrity. The acids can also combine with airborne particulate matter, heavy metals, and salts, leading to the development of black crusts. Since a major part of the college is constructed out of stones like basalt, limestone, siltstone etc, damages

[12] Ludwig, J. H., & Steigerwald,, B. J. (1965, July). Research in Air Pollution- Current Trends. *A.J.P.H*, *55*(7), 1028-1092.

caused by such pollutants have been documented. Air pollution also contributes to corrosion, rusting of metals and biodeterioration of structures.[13]

Risks to the structure

Cracks- Cracks have been spotted at an alarming rate in various parts of the college. Cracks are observed on walls, on concrete floors, on stairs, on mason pillars, on limestone tiles etc. The degree of cracks spotted range from superficial hairline cracks, shrinkage cracks, vertical cracks, floor cracks, structural cracks among others. Cracks disrupt the structural integrity of the structures and lead to long term maintenance issues. Appearance of cracks also suggests the aging of the structure.

Discoloration- Discoloration of painted walls, stone pillars, masonry walls have been spotted throughout the college. Not only does discoloration hamper with the aesthetic value of the structures but it also hints towards underlying structural error such as water seepage, weakening of wall membrane among other things which can crop up hazardous damages in long term if not mended.

Tenting of tiles- Tile tenting not only demands reworking of the base but also points towards the inherent weakness of the structure.

Biodeterioration or biodegradation- It is the undesirable change in the properties of a material caused by the vital activities of organisms. Bio-degradation has been spotted in structures on the terrace. Presence of surface layer of microorganisms and their products known as biofilms are defined as surface accumulations of organic products of biological biological activity.[14] These biofilms though do not affect the performance of material but so tamper with its aesthetic value. However, in case of physical bio deterioration, the organism distorts the material by growth or movement. Expansion of microbial masses between rock or stone layers can lead to spalling and hence affect the structural integrity.

Micro-vegetation- Microorganisms such as fungi and algae can be found growing on otherwise undamaged material utilizing surface dirt and detritus, but nevertheless detracting from the value

[13] Ludwig, J. H., & Steigerwald,, B. J. (1965, July). Research in Air Pollution- Current Trends. *A.J.P.H*, *55*(7), 1028-1092.
[14] Allsopp, D., Seal, K. J., & Gaylarde, C. C. (2004). *Introduction to Biodeterioration* (II ed.). Cambridge University Press. 2003060350

or acceptability of material. Evidence of which has been found on the terrace in the form of a hoard of neglected masonry capitals.

Magnitude of risk

The magnitude of a risk is measured in terms of its probability of occurrence and the severity of its consequences.[15]

The scale of severity measures how severe the consequences will be for each risk. There are five levels in the scale of severity.

- Negligible (1): The risk will have little consequences if it occurs.
- Minor (2): The consequences of the risk will be easy to manage.
- Moderate (3): The consequences of the risk will take time to mitigate.
- Major (4): The consequences of this risk will be significant and may cause long-term damage.
- Catastrophic (5): The consequences of this risk will be detrimental and may be hard to recover from.

The scale of likelihood identifies the probability of each risk occurring. There are 5 levels in scale of likelihood

- Very likely (5)
- Probable (4)
- Possible (3)
- Not likely (2)
- Very unlikely (1)

Scale of risk (likelihood*severity) is rated on a 1 through 25 scale

- Low (1-6): Low-risk events likely won't happen
- Medium (7-12): Medium-risk events can be prevented and mitigated, don't need to be top priority but shouldn't be ignored.
- High (13-25): these risks are likely to happen and have serious consequences.

[15] Preyssl, C., Atkins, R., & Deak, T. (1999, March). *Risk Management at ESA* [Bulletin 97].

Undesired outcome	Severity	Likelihood	Risk
Weakening/instability of structure due to floor/ wall cracks	Major	High likelihood	High
Weakening of wall membrane of the structure due to water seepage	Major	High likelihood	High
Instability of floor due to tenting of tiles/improper placement of floor tiles	Major	Moderate likelihood	High
Weakening of the structure due to physical biodeterioration	Moderate	Moderate likelihood	Medium
Negligence\abandonment of certain parts of the structure due to micro vegetal growth	Moderate	Moderate likelihood	Medium
Spalling and damage to structure due to human accidents	Moderate	High likelihood	High

CONSERVATION ACTION PLAN

Chapel

Investigation by Visual Inspection-

Visual inspections were carried out on 30 June 2023, 03 July 2023 to record defects observed at the Chapel. Inspections cover both the first and second floor of the Chapel. Types of damage detected frequently in the chapel are a) stains on walls and floor caused by water, inappropriate cleaning method and organic material, b) flaking of plaster, c) cracks on walls and window glasses, d) discoloration. Other types of damages encountered less frequently are bio-accumulation, broken edges, erosion, detachments, dirt build-up and paint deposits.

Agents of deterioration-

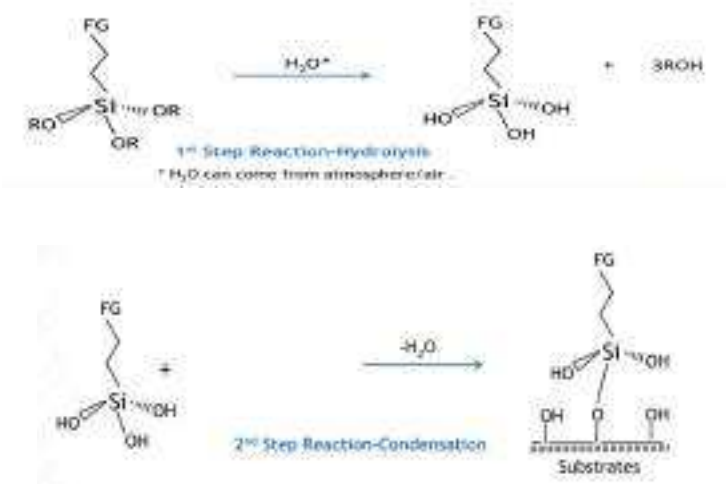
Water, physical forces (accidents), temperature, relative humidity, dust and dirt, human actions

Preventive conservation strategy-

- Application of a silane/siloxane based sealant to keep water from entering the walls.

Silanes and siloxanes, the two most common types of water repellents, are both derived from the silicone molecule. Silane/Siloxane chemically reacts with moisture in the substrate to form a hydrophobic, water repellent resin within the pores and on the surface. [16]
Both silane and siloxane are penetrating type sealers which allow the substrate to breath, allowing water vapor to escape while repelling water from outside. They also help to reduce mold and mildew growth, and efflorescence.[17]

[16] *SILANE AND SILOXANE WATER REPELLENTS FOR CONCRETE AND MASONRY*. (n.d.). The Euclid Chemical Company.
[17] Ibid



Reaction of silane with atmospheric moisture[18]

Chemical Compositions-

Silane-SiH4 Siloxane- H6OSi2

Suggested products- Sikagard-72 W; Siloxa-Tek 8500; Sure Klean Weather Seal.

- Bi-weekly cleaning of several elements of the chapel, such as floors, windows, door as well as furniture and ornaments. Thorough cleaning can be performed manually with the help of dry and wet mops to remove dust build up. A neutral pH balanced cleaner along with warm water can be used to clean floors, stairs.[19] Stained glass windows can be cleaned using an aerosol foam cleaner spray containing mild detergents and surfactants and for drying of stained glass a microfiber towel should be used.

  Suggested products- Agar pH-7 Neutral detergent, Betco pH-7 floor cleaner
  Sprayway Glass Cleaner Foaming Aerosol Spray

Remedial Conservation Strategy-

[18] Underhill, P. R. (2000, June). Effect of humidity on the curing of 3-glycidoxypropyltrimethoxy silane. *International Journal of Adhesion and Adhesives*, *20*(3), 195-199.
[19] *Acidity, Alkalinity, the pH Scale and How These Relate to Natural, Engineered, and Sintered Stone*. (n.d.). Lustro Italiano.

- Carry out a thorough cleaning of the door, walls, windows as well as furniture and ornaments. Flat mops can be used to remove dust build up and stains. A neutral pH balanced cleaner along with warm water can be used to clean floors, stairs. Window frames can be cleaned using water and non-alkaline detergent with pH-7, to prevent any damage to the surface of the frame. Stained glass windows can be cleaned using an aerosol foam cleaner spray containing mild detergents and surfactants. Microfiber towels to be used to clean and dry stained glass, they can also remove dust, first and other residues from glass.

  Suggested Products-Agar pH-7 Neutral detergent, Betco pH-7 floor cleaner
  Sprayway Glass Cleaner Foaming Aerosol Spray

- Bi-annual repair or installation of new stained window glasses.

- After fixing the source of leakage via application of sealant, the walls should be repainted with a waterproof and stain blocking paint. A stain blocking paint allows the damp area to breathe and dry out whilst covering it and sealing the damp from spreading any further, darkening or smelling worse.[20] In case of water stains on masonry walls, poultice can be used. Poultice is a liquid cleaner or chemical mixed with a white absorbent material to form a paste which is applied to the stained area about ½ inch thick. The poultice is then covered with a plastic, the drying process (which lasts till 24 to 48 hours) pulls the stain out of the stone and into the poultice material. Distilled water is then used to clean the wall.[21] In case of deep water stains on masonry walls, 000 or 0000 steel wool can be used to buff the stain.[22]

  Chemical composition of poultice- kaolin ($Al_2Si_2O_5(OH)_{4)}$, fuller's earth (SiO_2 55%; $A1_2O_3$ 6%; CaO 3.5%; MgO 2.0%, Fe_2O_3 6%; Water 10% representing montmorillonite 50% and silica 18%.)[23], Whiting (CaCO3, MgCO3, CaMg (CO3)2)[24], earth, powdered chalk (CaCO3), white molding plaster (CaSO4. 1/2H2O) or talc ($Mg_3Si_4O_{10}(OH)_2$)[25]

[20] *WO2005030495A2 - Methods of blocking stains on a substrate to be painted, and composites suitable for use in such methods*. (n.d.). Google Patents.
[21] *Remove Stains from Stone Applications*. (n.d.). Natural Stone Institute.
[22] Ibid.
[23] *Fuller's Earth - Pharmacognosy*. (n.d.). Pharmacy 180.
[24] *Whiting Powder Manufacturer,Exporter,Supplier from Maharashtra*. (n.d.). Welcome Chemicals
[25] *Remove Stains from Stone Applications*. (n.d.). Natural Stone Institute.

Suggested products-

WEpaint anti staining paint, Dulux Superclean 3 in 1 Stain Proof Paint
Sealers plus poultice powder
Metalino Steel Wool 000 EXTRA FINE

- Cracks in plaster can be mended via re-plastering or using plaster filler like polyfilla or a powder filler for filling plaster cracks.

  Chemical composition of polyfilla[26] - Talc ($Mg_3Si_4O_{10}(OH)_2$); Bisphenol A-(epichlorohydrin); Di- "isodecyl" phthalate; Titanium dioxide (TiO2); 1,4,6,-tris (dimethylaminomethyl)phenol

  Suggested products- Polyfilla masonry patching plaster

- Dirt accumulation on wooden doors can be removed using warm water to wash the durst away, followed by drying of treated area using a microfibre towel.

Expansion Building

Investigation by Visual Inspection-

Visual inspections were carried out on 03 July 2023 to record defects observed on the 1st, 2nd and 3rd floor of the expansion building. Types of damage detected frequently are- a) floor concrete with gaps and cracks running along the length b) spalling of concrete c) cracks on walls, and d) lifting of floor tiles and e) dirt build up. Other types of damages encountered less frequently are rusting, creation of isolated beehives.

Agents of deterioration-

Aging of structure, poor workmanship, movement of concrete due to variations in humidity levels and temperature, dirt build up

Preventive conservation strategy-

[26] *SAFETY DATA SHEET*. (n.d.). SAFETY DATA SHEET.

- Frequent cleaning of floors, stairs and doors to prevent dust build up. Tile floors need to be swept with the help of a vacuum attachment with soft bristles or a soft-bristle broom to remove dirt, dust, hair and debris. A neutral pH balanced cleanser along with warm water can be used to clean floors. Wooden doors can be cleaned using mild detergent and water and a non abrasive scrub to clean the surface. A microfibre towel needs to be used to dry off the wooden door complementary after cleaning.

  Suggested products- Agar pH-7 Neutral detergent, Betco pH-7 floor cleaner
  NC-912 NEUTRAL pH CLEANER (for doors)

- Bi-annual inspection and early spotting of cracks on plaster, walls and floor

Remedial conservation strategy-

- Narrow concrete cracks can be filled using masonry crack elastomeric filler that can be applied with a caulking gun.[27] Alternatively, many products come in bottles with an applicator tip. In case of wide concrete gaps, the crack should be troweled with a concrete patching compound.[28] Before filling the cracks, it is essential to clean any loose material, debris from the old concrete using a wet/dry shop vacuum or brush.[29] Also the trowel should be stabbed into the compound to prevent air pockets and patch the material deep into the crack.

- Daily cleaning of floors using a dry mop and a neutral pH balanced cleaner. Wooden doors can be cleaned using mild- detergent and water and a non abrasive scrub to clean the surface, followed by drying of the door with the help of a microfibre towel.

  Suggested products- Agar pH-7 Neutral detergent, Betco pH-7 floor cleaner
  NC-912 NEUTRAL pH CLEANER (for doors)

- Cracks in plaster can be mended via Re-plastering or using plaster filler like polyfilla or a powder filler for filling plaster cracks.

[27] *How to Repair Wide Cracks in Concrete*. (2023, June 2). The Spruce.
[28] Ibid.
[29] Ibid.

Chemical composition of polyfilla[30] - Talc ($Mg_3Si_4O_{10}(OH)_2$); Bisphenol A-(epichlorohydrin); Di- "isodecyl" phthalate; Titanium dioxide (TiO2); 1,4,6,-tris (dimethylaminomethyl)phenol

- Water based rust removers can be used in substitute of strong acidic or alkali rust removal solutions to prevent harm to the conservator-persevere and to the object being treated. Water based rust removers draw the rust away or lift it from the metal via a process known as chelation.[31] Aforementioned process causes molecules in the rust removal solution to bond with the rust and draw it away from the undertaking metal into a substrate. These rust removers also come in gel form.[32]

  Chemical composition of rust remover -Water Soluble Rust attacking acid selected from the group consisting of Phosphoric acid (H_3PO_4), Citric Acid ($C_6H_8O_{7)}$ and Trichloroacetic Acid (C2HCl3O2), Water, ⅓-5% based on the weight of the water and acid, of Sodium Carboxymethyl Cellulose (C8H15NaO8) a gel forming agent and 3-30% based on the Pectin (C6H10O7) a gel stabilizer[33]

- Improper or poor application of tile adhesive can lead to tenting of tiles caused by stress due to the continuous (and natural) expansion and contraction of the flooring system. An isolated case of Tile tenting has been observed occurring near the department of statistics. It indicates that the tile adhesive used was either weak or it was applied poorly.[34] Since the section of tented tile is small, it can be repaired by releasing the pressure between the tiles and putting them back into place using a flexible polymer modified adhesive and replacing the surrounding grout.[35] A flexible polymer modified adhesive, like the European C2 tile adhesive will increase bond strength and flexibility of the floor system to handle high stress.[36]

  Suggested products- European C2 tile adhesive

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[30] *SAFETY DATA SHEET*. (n.d.), op. cit
[31] Ferrer, K. S., & Kelly, R. G. (2001, February). Comparison of Methods for Removal of Corrosion Product from AA2024-T3. *Corrosion*, *57*(2), 110-117. 250394590
[32] Ibid.
[33] *US2558167A - Rust remover*. (n.d.). Google Patents.
[34] *CAUSES OF TENTING*. (n.d.). Ezee Tile Adhesives.
[35] Ibid
[36] Ibid.

Terrace

Investigation by visual inspection was carried out on 04 July 2023 to record defects observed at the Terrace. Inspections cover the Hindi Department, Washroom building, MVM Office and a hoard of discarded stonework. Types of damage frequently detected are a) Biodeterioration b)micro and macro vegetation c) water stains on walls, pillars and doors, d) missing or lost elements of concerned structures/ detachments, e) discoloration due to dirt build-up. Other types of damages encountered are paint deposits, scratches, hairline cracks on walls and pillars, bird excreta

Agents of deterioration-

Damp environment, pollution, Rh-temperature, light

It was observed that on days marking heavy rains, the condition of structures on the terrace was alarming as moisture accelerates biodeterioration and promotes growth of micro vegetation, leading to dirt build up and soiling. However on dry days, the condition of structures seemed to improve with considerable reduction in soiling and dirt build up on the structures.

Preventive conservation strategy-

- Since the structures are located on the terrace, they are vulnerable to both humidity which favors growth of algae, fungi and light which is necessary for photosynthetic organisms to thrive, such as lichens and mosses. Early spotting of biodeterioration accompanied with frequent cleaning of the structures using tools such as air abrasions, scrapers etc to prevent build up of organic material.

Remedial conservation strategy-

- Removing micro vegetation/ surface growths/ biofilms -the most economic and effective method would be using mechanical methods removing biodetriogens on structures using tools such as brushes, scalpels, spatulas, scrapers, air abrasives etc. Phototrophic biofilms should be completely dry before any mechanical cleaning, because dry crusts often

readily detach from the materials and can be easily removed by using brushes, or low-pressure washing.[37]

- Since the structures are in an open environment, rusting of iron objects is a common phenomenon. Oxalic acid can be used to remove severe rusting.

  Chemical composition of Oxalic Acid- $C_2H_2O_4$[38]

- Replastering and repainting of pillars and MVM and the Hindi Department buildings using a stain blocking paint.[39] Aforementioned paint allows the damp area to breathe and dry out whilst covering it and sealing the damp area from spreading any further, darkening or smelling worse.

Restorative conservative strategy-

- Almost all structures and pillars found on the terrace are missing their top most component or their apex/ capital. Correspondingly a hoard of neglected stonework, pillar tops/ capitals (which now serves as breeding ground for funaria, algae and moss) can be found on the terrace. Aforementioned pieces might be the displaced components of the still-standing pillars. Reason behind removal of these components/apexes remains unearthed, degradation of the capitals can be cited as one among many hypothetical rationale. In order to restore the aesthetic value of the structures, the pillars need to be restored to their original form, with installation of new capitals.

First Quadrangle

Investigation by visual inspection was carried out on 04 July 2023 to record defects observed in the Front Quad. Types of damage detected frequently are a) cracks in structures b) water leakage and water stains c) chipping off plaster and c) broken tiles. Other damages documents are creation of an isolated beehive, soiling.

Causes of deterioration-

[37] Capitelli, F., Catto, C., & Villa, F. (2020, October). The Control of Cultural Heritage Microbial Deterioration. *PubMed central*, *8*(10).
[38] *Oxalic acid | Formula, Uses, & Facts*. (n.d.). Britannica.
[39] *WO2005030495A2 - Methods of blocking stains on a substrate to be painted, and composites suitable for use in such methods*. (n.d.). Google Patents.

Accidents, water, Rh-temperature

Preventive conservation-

- Frequent Cleaning of floor and pillars using a neutral pH balanced cleaner to prevent dirt build up and soiling.

- Masonry pillars are generally porous and during rain events water is absorbed into them splashed onto them and reabsorbed into them from the ground. Water repellents can prevent ingress of water and prevent decay of stone. Water repellency can be provided largely by alkoxysilanes, silicones, and fluo-ropolymers.[40]

Remedial conservation measures-

- Repair and repainting of broken edges of the entrance arch with help of cement, followed by a white wash and paint.

  Chemical composition of cement- Portland cement is made up of four main compounds: tricalcium silicate ($3CaO \cdot SiO2$), dicalcium silicate ($2CaO \cdot SiO2$), tricalcium aluminate ($3CaO \cdot Al2O3$), and a tetra-calcium aluminoferrite ($4CaO \cdot Al2O3Fe2O3$).[41]

- Cleaning of water spots and rings can be done with the help of 000 or 0000 steel wool.[42]

- Slight surface scratches can be buffed with dry 000 or 0000 dry wool. In case of deeper scratches and nicks re-polishing the pillar can be considered.[43]

Restorative measures

- An isolated portion of damaged limestone tiles along the Quad boundary outside the seminar needs to be extracted and reinstalled with fresh limestone tiles.

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[40] Price, C. A. (1996). *Stone conservation : an overview of current research*. Getty Conservation Institute.
[41] *Cement - Composition, Properties, Major Cements*. (n.d.). Britannica.
[42] *Remove Stains from Stone Applications*. (n.d.). Natural Stone Institute.
[43] Ibid.

ANALYSIS

A Conservation Action Plan is the first step to any conservation measures undertaken. Firstly, we conducted the photographic documentation of the damaged area which showed cracking of the wall, rusting of iron rods in the wall, micro vegetational growth, water leakage and tenting of floor tiles.[44] Visible cracks on the wall caused by water leakage and moisture can also ruin adjacent walls by increasing the moisture content and eventually formation of more structural cracks. Weathering must be considered as a baseline for the micro-vegetational growth, lichens are a symbiotic association of photosynthetic microorganisms geld in a mass of fungal hyphae, with growth in various shrublike (fruticose), leaf-like (foliose) or encrusting (crustose) forms. Tenting of floor tiles may be caused by unsuitable cleaning that does not support the uniform bonding layer. The next cause is the unevenly set tile.

Furthermore, we utilized several advanced tools such as distometers, Dino lights, UV lights, compasses, Epicollect5 application and Google Earth for the examination and documentation of St. Xavier's College, Mumbai. We used varying methods for documentation that included visual inspection based on eye level , Visual inspection based on lights and radiation , Measuring tape, Monitoring equipment and softwares. Following this, we mapped the area where research would be conducted. The aerial Maps shown above of the 1st Quadrangle and Terrace. With the help of a collaborative approach with the geology department we were able to analyze and document the material and technology used in the construction of St. Xavier's Colleges (Basalt , Rhyolite , Limestone, Mortar, Brick, Gypsum Plaster, Metal, Wood, and Stained Glass). The outcome of the careful examination and documentation was a table of the glossary of damages of Chapel, Terrace, Expansion Joint and 1st Quadrangle. For instance, in the Chapel, we observed the accumulation of dirt, effects of Rh and Temperature resulting in discoloration and damp wall spots that caused flaking of plaster coatings. In the terrace, we examined the biodeterioration caused by water, rh and temperature, light exposure and air pollution.

Similarly in the expansion joint area, we also deeply examined the cracks, impaired cement, concrete floor, cracked wall, rusting, a beehive and tenting of floor tiles. All point to the occurrence of these major factors of deteriorations viz. Rh-temperature, faulty attempts of

[44] Mulla, Fayaz. (2018). "Study on Causes & Control of Cracks in A Structure Building Construction". IJSART. Vol. 4, No. 12. pp. 409-412.

conservation and external forces. Due to the occurrence of vegetational growth in the terrace walls of the MVM office and washroom cubicle, cracks have developed throughout the structure. The roots of the growing trees penetrate within the foundation pillars and walls, further worsening these cracks. Timely precaution helps combat these severe cracking issues. In the 1st Quadrangle, the water accumulated in the gargoyles drips during monsoons causing water leakage and chipping of paint. Broken limestone tiles on the edge of the Quadrangle caused by negligence and human error. Some of the future steps that have to be taken are installation of central utility plants, especially the ones with high efficiency, environmentally friendly central plant technology that makes college premises eco-friendly.

A Conservation Action Plan aims to preserve and restore historic structures, ensuring longevity for future generations. With the use of appropriate techniques and instruments we analyzed the heritage structure deeply and clearly, documenting the damages and developing techniques for the risk assessment and future conservation.

CONCLUSION

Heritage structures are a symbol of cultural identity. Social, cultural, historical significance of any structure can be visualized through the architecture style, design, role or connection of the structure with any significant political or historical event etc. St. Xavier's College has been recognized as a heritage structure by the Mumbai Heritage Conservation Committee.[45] This paper outlines and recommends various conservation practices that can be adopted at the aforementioned heritage structure.

Conservation is an activity taken to prolong the life of historic structures. Before embarking on any physical intervention or design strategy for a potential conservation project it is essential to assemble all the available evidence on a building or site that is to be conserved. Historical evidence such as old pictures, maps and reports from previous investigation of any historical reports can inform and enhance current conservation practices. Historical research has to be conducted well in advance of physical investigation.[46] Measured drawings and maps of the structure highlighting the present metrics and condition of the structure are prepared thereafter.

[45] *Self Study Report for 4th Cycle of Accreditation* [Submitted to National Assignment and Accreditation Council, Banglore]. (2021, September).
[46] Harun, S. N. (2011). Heritage Building Conservation in Malaysia: Experience and Challenges. *Procedia Engineering*, *20*, 41-53.

The historical information and pictures of the site under consideration were sourced via the college magazines published by the college board itself. Aforementioned source provided a chronological outline of various developments which happened to the college structure over the course of time. Maps of the current college structure, particularly maps of the terrace and the first quadrangular of the college campus, were created with aid of 'google maps'.

After mapping of the structure, a survey is performed to aid in identifying and recording building defects through means of photographic and digital documentation The core of the survey is the process of walking through the structure to gather and record information based on observation of the finished surfaces or any exposed surfaces, to understand the state of defects and determine causes behind it.[47] The survey demands a multidisciplinary approach, for example laboratory testing aids in identifying the original material and the composition of materials used in the structure.[48]

Survey of the terrace, first quadrangular, chapel and expansion building situated in the college campus revealed a multitude of damages. The photographic documentation of the damages spotted at various areas in the college allowed the creation of a 'glossary of damages'. The aforementioned glossary not only helps in creation of a condition report of the structure but also forms the base of the recommended conservation plan for the heritage structure.

After diagnosing the factors of decay and damages to the structure, a proposal or method of statement on techniques to repair and conserve the structure is outlined. The plan includes various conservative actions providing aspects of preventive, remedial and restorative conservation. The plan requires approval by relevant institutions such as the college committee or the stakeholders of the college. The college authority has to manage and monitor the total product and perform related duties such as design, planning, monitoring, arrangement and finding funds for successful implementation of the conservation plan.

The case study of St. Xavier's College highlights the imperative of protection, conservation, preservation, and restoration efforts within educational institutions.

[47] Harun, S. N. (2011). Heritage Building Conservation in Malaysia: Experience and Challenges. *Procedia Engineering*, *20*, 41-53.
[48] Ibid.

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Plates



Plate 1. Location of St. Xavier's College

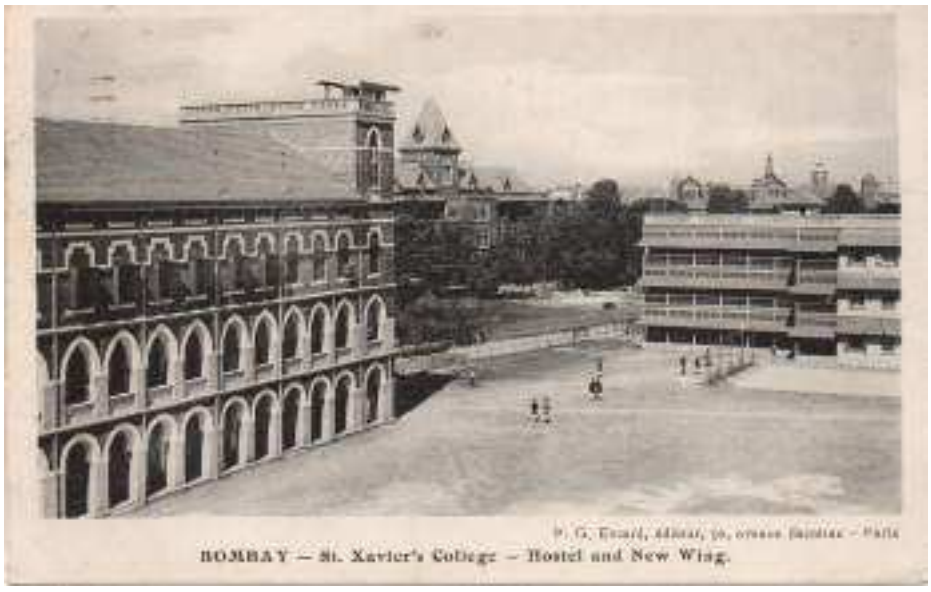


Plate 2. St. Xavier's College Hostel and New Wing

Inspection of samples under Dino lite (digital microscope)



Plate 3. Brick sample collected from the terrace



Plate 4. Mold sample collected from terrace (MVM room)

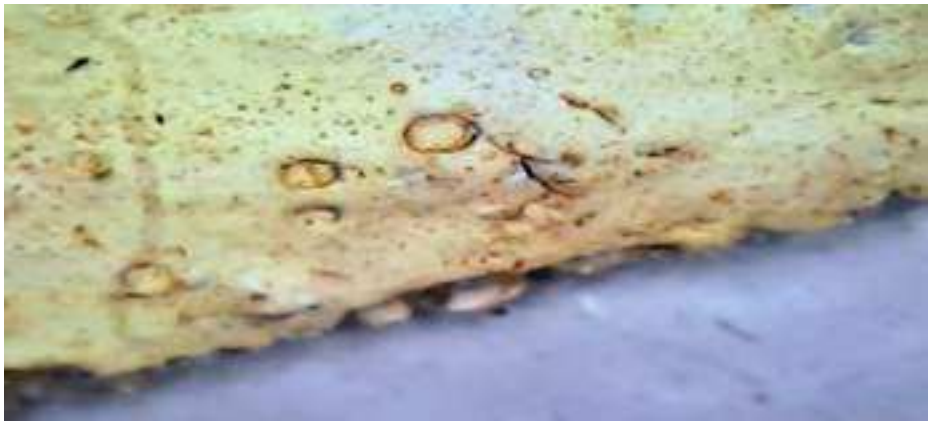


Plate 5. Plaster sample collected in front of the Seminar room.



Plate 6. Concrete sample with vegetation from the terrace.

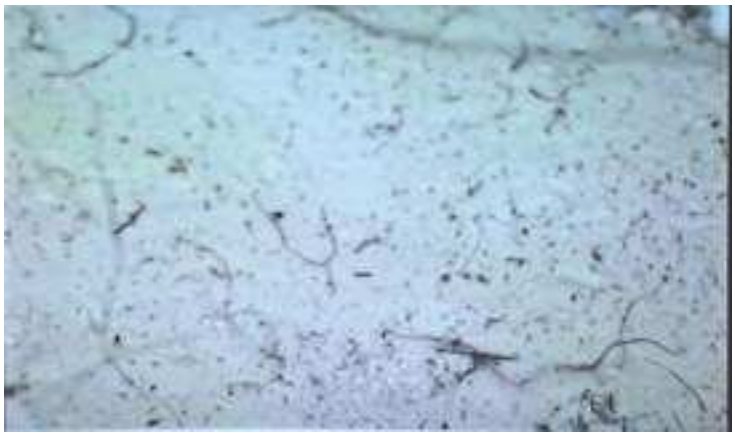


Plate 7. Paint sample collected in front of the Seminar room.

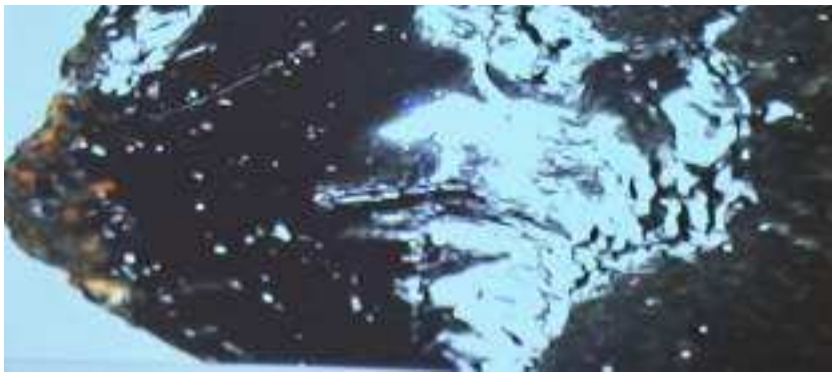


Plate 8. Tar sample collected from the heap of stones beside the Solar inverter room on the terrace.

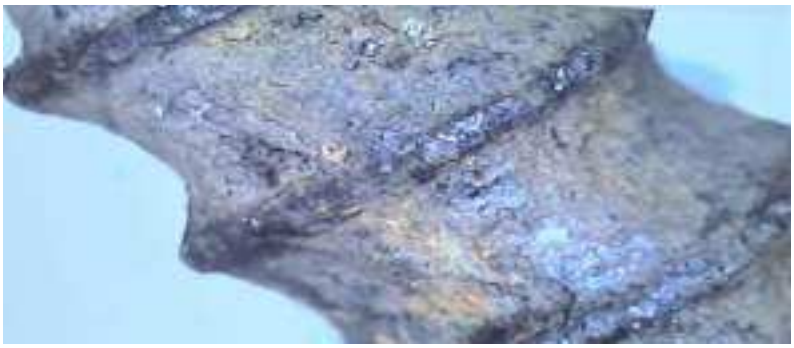


Plate 9. Iron nail collected from the heap of stones beside the Solar inverter room on the terrace.



Plate 10. Conservation Treatment Undertaken in the past of the Chapel

Images of samples





Sample 1 Limestone Tiles [Limestone Tile tested with Hydrochloric Acid (HCL)]



Sample 2 (Plaster)



Sample 3 (Plaster and Paint)



Sample 4 (Cement)



Sample 5 (Mold)



Sample 6 (Moss)



Sample 7 (Paint)



Sample 8 (Nail)



Sample 9 (Stone)



Sample 10 (Tar)



Sample 11 (Brick)

Group examination and documentation



The group documenting and collecting samples from the Expansion joint (LR23).



Examination and Documentation of Chapel 1st floor.



Sample collection on the Terrace near the MVM (Marathi Vandmay Mandal) room.



Team identifying the different materials used during the construction of the college.



Identifying damages in the pillar in 1st Quadrangle



Team at the 1st Quadrangle